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$$\int \frac{x-1}{x^2} \ln x dx = \int \left(\frac{x}{x^2} - \frac{1}{x^2} \right) \ln x dx = \int \frac{\ln x}{x} dx - \int \frac{\ln x}{x^2} dx$$

Eerste integraal:

$$u = \ln x \Rightarrow du = \frac{1}{x} dx:$$

$$\int u du = \frac{u^2}{2} = \frac{\ln^2 x}{2}$$

Tweede integraal:

$$\int \ln x \cdot x^{-2} dx$$

$$\begin{cases} u = \ln x \\ dv = x^{-2} dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{x} dx \\ v = -\frac{1}{x} \end{cases}$$

$$\int \frac{\ln x}{x^2} dx = \ln x \cdot \left(-\frac{1}{x} \right) - \int \left(-\frac{1}{x} \right) \cdot \frac{1}{x} dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x}$$

Dus:

$$\int \frac{x-1}{x^2} \ln x dx = \frac{\ln^2 x}{2} - \left(-\frac{\ln x}{x} - \frac{1}{x} \right) + c = \frac{\ln^2 x}{2} + \frac{\ln x}{x} + \frac{1}{x} + c$$

References